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Machine Learning based Epileptic Seizure Detection from EEG Signal: Bangladesh Perspective

Shupta Das
Dept. of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna -9203, Bangladesh
shuptadas25203@gmail.com

Suraiya Akter Mumu
Dept. of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna -9203, Bangladesh
suraiyaaktermumu26@gmail.com

Abdus Salam
Shaheed Sheikh Abu Naser Specialized Hospital & Prince Hospital
Khulna, Bangladesh
a.salamk51@gmail.com

M. A. H. Akhand
Dept. of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna -9203, Bangladesh
akhand@cse.kuet.ac.bd

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***Abstract*—Epileptic seizures (ES) pose significant challenges to individuals' well-being and quality of life, with a global impact affecting millions of people. Detecting and classifying ES is essential for proper medical intervention and management. This study addresses ES detection using EEG analysis, focusing specifically on the context of Bangladesh. This study presents a EEG-based method for detecting epileptic seizures differentiating between seizure and non-seizure signals. Utilizing empirical mode decomposition, essential subcomponents are extracted from EEG signals, from which three distinct features were derived. Neural Network (NN), the well-known machine learning model, is employed for ES classification from the extracted features. The significance of this research lies in the utilization of real-life EEG data from a hospital in Khulna, Bangladesh, rather than relying solely on publicly available benchmark datasets. Processing real-life EEG data presents unique challenges, including data preprocessing and variability in EEG signals due to different devices and channel configurations. The proposed model achieved remarkable average accuracies, with 91.05% using 50 hidden neurons in the NN. To validate the effectiveness of the proposed method, outcomes were also compared with publicly available benchmark dataset. The proposed method offers a reliable and efficient approach for ES detection, particularly relevant for real-life healthcare settings in Bangladesh, contributing to improved epilepsy management.**

***Keywords*— Epilepsy, Seizure, EEG, Empirical Mode Decomposition, Neural Network**

I. INTRODUCTION

Epileptic seizures (ES) have significant impacts on a person's quality of life which are caused by a disruption in the normal electrical activity of the brain. The impact of ES is around 50 million individuals worldwide [1], and in Bangladesh, there is an estimated occurrence of 8.4 reported cases per 1000 people [2] (with a 95% confidence interval). By understanding and addressing ES, medical professionals can help individuals with epilepsy manage their condition and to cure or reduce the frequency and severity of seizures. ES detection can be challenging due to the unpredictable nature of seizures and the variability in their presentation among individuals. Additionally, seizures may sometimes manifest with subtle or atypical symptoms, making them difficult to identify through traditional observation alone.

As the root cause of epileptic seizures (ES) is a neurobiological brain disorder, detection through analyzing brain signals is a favorable way. Among different brain signals, Electroencephalogram (EEG) is considered one of the easiest and widely used methods because it is non-invasive, relatively inexpensive, and provides real-time information about brain activity [3]. Therefore, in the field of epilepsy research, various approaches is employed to detect and classify seizures using EEG signals. By leveraging modern machine learning (ML) techniques, seizure detection can be automatically processed and analyzed from continuous EEG data.

The most of existing ML-based studies are with CHB-MIT EEG dataset. One study [4] focused on the Tunable Q-wavelet transform(TQWT) based decomposition technique combined with feature extraction, including nonlinear, temporal, and statistical features. The classification task was carried out using Support Vector Machine (SVM) and Random Forest (RF) classifiers. In another investigation [5], two distinct time domain feature extraction methods were applied, and various classifiers like SVM, K Nearest Neighbors (KNN), Logistic Regression (LR), RF, Decision Tree (DT), and Naive Bayes (NB) were tested for seizure detection. In a separate approach [6], researchers delved into the use of Mutual Information (MI) to identify correlations between channels. They also employed Convolutional Neural Networks (CNN), specifically 2D CNN, to directly detect seizures without the need for explicit feature extraction. Furthermore, a different study [7] focused solely on using 2D CNN to detect seizures without any prior feature extraction. In [8], researchers utilized Graph Attention Networks (GAT) to extract spatial features, and Bi-directional Long Short-Term Memory (BiLSTM) was employed for ES detection directly from raw EEG signals.

In the context of epilepsy research, the Bonn dataset is another widely used publicly available dataset [9]. In one study [10], researchers employed a Feed Forward Neural Network (NN) for classification. They utilized wavelet decomposition in conjunction with feature extraction techniques to prepare the data for the classification task. Another study [11] focused on refining the parameters of a hybrid Support Vector Machine (SVM) using Genetic Algorithm and Particle Swarm Optimization methods.

The aim of the study is ML-based ES detection using real-life EEG data from a hospital located in Khulna, Bangladesh. The publicly available benchmark EEG datasets (e.g., CHB-MIT) are already pre-processed; therefore, existing studies did suffer from the pre-processing hassle which is a great concern to work with real-life EEG data. Moreover, EEG signal depends on device used and number of channels. In real-life aspect of Bangladesh, standard ML-based ES detection has been investigated in this study from raw data preprocessing to classification of ES. Empirical mode decomposition is used to decompose the EEG signals into subcomponents, and three specific features were extracted from the subcomponents. These features were then used as inputs for a NN classifier to distinguish between seizure and non-seizure states. To validate the effectiveness of the proposed method, outcomes is also compared with CHB-MIT dataset.

The rest of the paper is organized as follows. Section II describes the methodology; section III describes the experimental results and section IV finally conclude the overall procedure with future directions.

II. Methodology

ES detection using EEG data from local hospital is the main concern of the study. Fig. 1 illustrates ES detection steps: Data Collection and Preprocessing, Signal decomposition, Feature Generation, and Classification. The following subsections describe individual steps.

A. EEG Data Collection and Preprocessing

The Prince Hospital Khulna, Bangladesh is the source of the EEG dataset used in this study and called hereafter as Prince Hospital Khulna (PHK) dataset. EEG signal is recorded under the supervision of a specialist on neurology and epilepsy, (i.e., an author of the study). The data is recorded using the Nicolet vEEG System, manufactured by Natus Neuro, USA. The dataset is in .e (Nicolet) format, with a sampling rate of 500 samples per second. The dataset holds continuous signal values of 19 channels including "FP1-A1",

TABLE I. Summary Of Patients Of PHK Dataset

Patient No.	1	2	3	4	5	6	7	8	9	10
Age (Year)	28	9	13	6.5	18	4	3	7	17	1.5
Gender	F	M	F	M	M	M	F	F	M	M

"F3-A1", "C3-A1", "P3-A1", "O1-A1", "FP2-A2", "F4-A2", "C4-A2", "P4-A2", "O2-A2", "F7-A1", "T3-A1", "T5-A1", "F8-A2", "T4-A2", "T6-A2", "FZ-A1", "CZ-A1" and "PZ-A1" in monopolar or referential montage. Bipolar montage settings (Longitudinal & Transverse) were also interpreted.

EEG data from 10 individual patients is considered in this study. Table I provides a concise summary of the patients in age and gender; the names, addresses, and other personal information of the patients were kept anonymous for privacy purposes. All the patients' data is analyzed, emphasizing the importance of investigating the overlapping characteristics between seizure and non-seizure data in individuals with

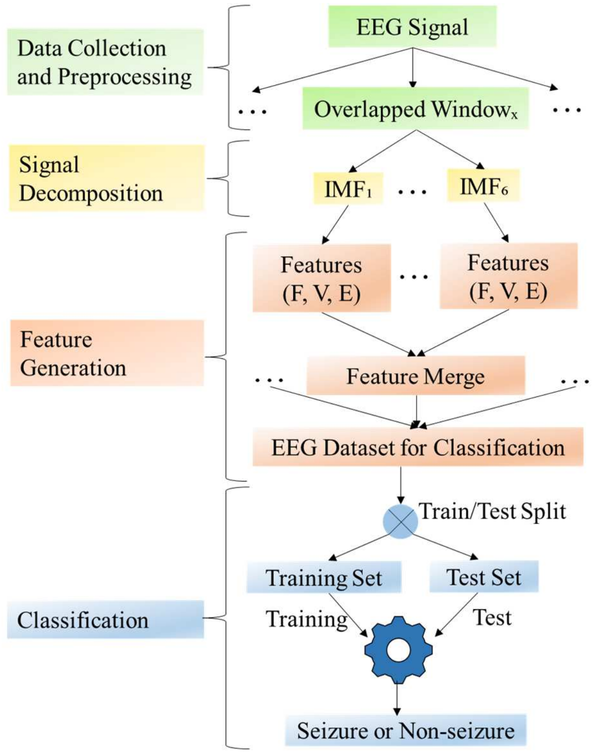

Fig. 1. Proposed Framework for Epileptic Seizure Detection.

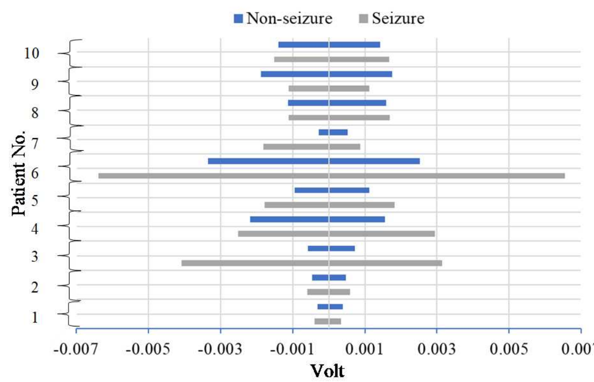

Fig. 2. Range of EEG signals in Volt of Individual Patients.

epilepsy. EEG data is electrical voltage collected from skull; volt ranges for both seizure and non-seizure activity were visually represented in the chart shown in Fig. 2. After collecting the Nicolet files, we utilized MATLAB to convert them into CSV files. The original sampling rate of 500 Hz is resampled to 256 Hz. From the dataset consisting of 10 cases, we carefully separated the seizure and non-seizure segments.

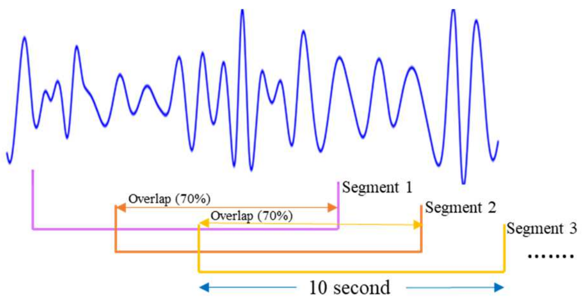

Fig. 3. Segmentation Process.

To address potential class imbalance issues, we ensure that the non-seizure portions is matched in duration with the seizure segments. Eventually, we select a total of 1,156 seconds of data for analysis. Each signal is then segmented into 10-second intervals with a 7-second overlap, as illustrated in Fig. 3. This process results in a collection of 380 segments, with an equal representation of seizure and non-seizure instances, creating a balanced dataset for further investigation and analysis.

For better understanding, publicly available CHB-MIT benchmark dataset is also considered in this study[12] .CHB-MIT benchmark dataset consists of EEG recordings from 23 subjects at Children's Hospital Boston. These subjects, comprising 5 males aged 3-22 years and 17 females aged 1.5-19 years, is divided into 24 cases. From 23 patients a total number of 7414 segments (3707 segments for seizure and 3707 segments for non-seizure) is extracted for analysis.

B. Signal Decomposition

EEG signal is decomposed using Empirical Mode Decomposition (EMD) technique. EMD is an adaptive and data-driven method that breaks down a signal into intrinsic mode functions (IMFs) based on local characteristics. The EMD process involves identifying local extrema in the signal, creating upper and lower envelopes connecting these extrema, and calculating the mean envelope as a reference for extracting the first IMF. This iterative process is then applied to the residual signal, obtained by subtracting the previously extracted IMF from the original signal, to extract subsequent IMFs [13]. In this study, a total of six IMFs from the EEG signals are considered as of existing study [3]. So, the equation stands:

$$x(t) = \sum_{n=1}^{6} IMF_n + r(t) \quad (1)$$

Here x(t) is original signal, r(t) is the residue. Sample IMFs are given from PHK dataset in Fig. 4.

C. Feature Generation

Feature extraction as an important step in the process of classification. From each IMF, three features are extracted. A total number of 18 features are extracted from six IMFs. A short definition of three features is given below and they are measured following the equations from [3] :

- *Fluctuation index: The Fluctuation Index is a measure that captures the relative magnitude of variability within a dataset. It is particularly useful for comparing multiple time series and uncovering patterns and trends inherent in the data.*
- *Variance: Variance is widely used in statistics and data analysis to assess the dispersion of data and understand the level of variability in a dataset.*
- *Ellipse area of SODP: The SODP is a graphical tool used in time series analysis to identify nonlinearity or*

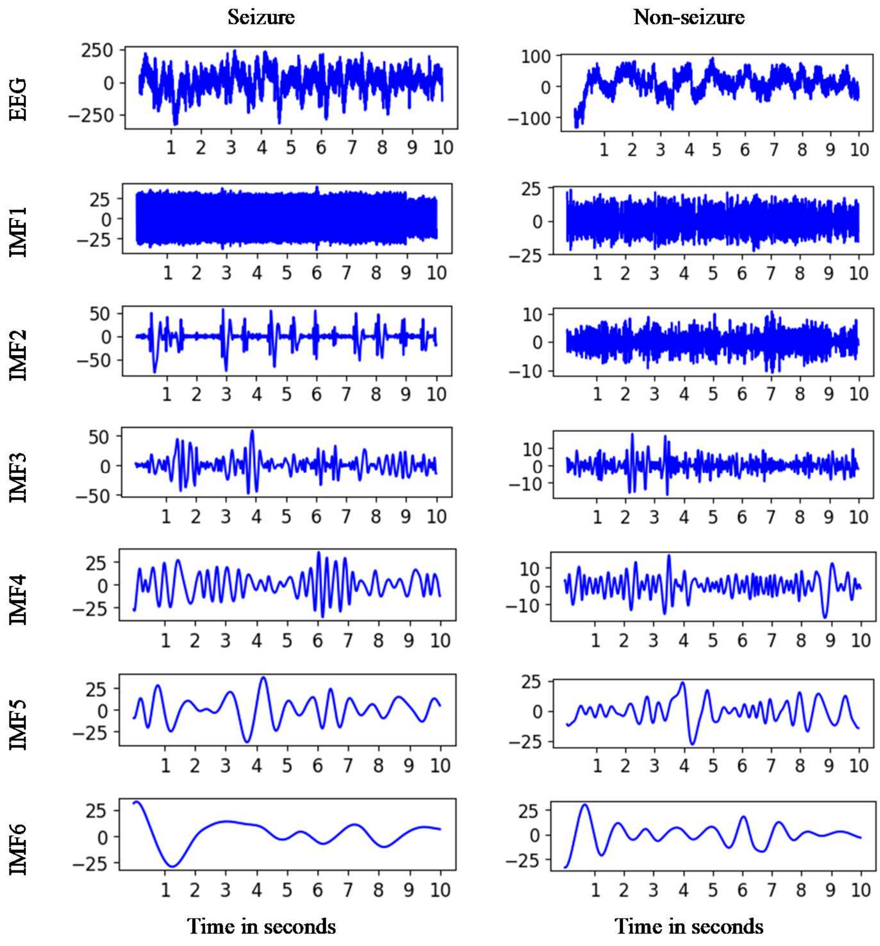

Fig. 4. Sample IMFs of PHK Dataset.

higher-order patterns in data. It involves plotting the second-order differences of the time series data against the data itself. The resulting plot often exhibits an elliptical or circular shape when certain patterns are present. the term "ellipse area" typically refers to the area enclosed by the elliptical shape formed by the data points on the SODP plot.

D. Classification using Neural Network (NN)

NN is the well-known ML method for intelligent classification task. For ES detection, a NN with specific configurations is considered to classify Seizure and Non-Seizure samples. The NN consists of three layers: the input layer, the hidden layer, and the output layer. For the CHB-MIT dataset, the input layer comprises 396 nodes, while for the PHK dataset, it contains 342 nodes. The hidden layer comes in two variations, each with a different number of neurons: one variation has 10 neurons, and the other has 50 neurons. The output layer contains 2 neurons, which are responsible for generating the final classification results with a Softmax activation function. The NN architecture included an activation function of *ReLU*, a learning rate of 0.001 and the *Adam* optimization algorithm. Here, *categorical_crossentropy* is the loss function and *accuracy* are the evaluation metric.

TABLE II. EXPERIMENTAL RESULTS OF CHB-MIT DATASET AND PHK DATASET

Dataset	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg.
CHB-MIT	NN(HN=10)	99.06	99.39	99.19	98.85	98.99	99.09
	NN(HN=50)	99.53	99.46	99.39	99.33	99.26	**99.39**
PHK	NN(HN=10)	92.11	88.16	84.21	93.42	96.05	90.79
	NN(HN=50)	90.79	90.79	85.53	92.11	96.05	**91.05**

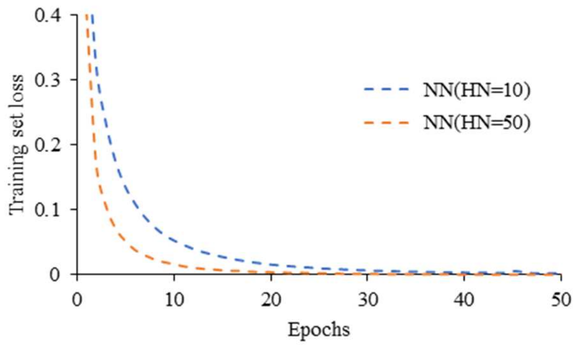

Fig. 5. Training Set Loss Vs. Epoch Curve for a Fold of CHB-MIT Dataset.

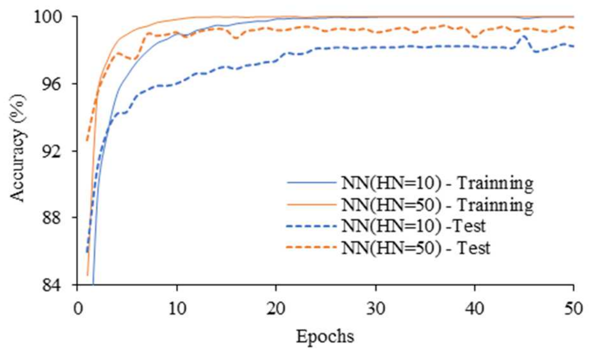

Fig. 6. Accuracy Vs. Epoch Curve for a Fold of CHB-MIT Dataset.

III. EXPERIMENTAL RESULTS

In the study, a classification task is conducted using a 5-fold cross-validation approach for both the PHK and CHB-MIT datasets which is demonstrated in Table II. Initially, the classification experiments are conducted on the CHB-MIT dataset. The loss curves are shown in Fig. 5 and the accuracy curves are shown in Fig. 6. The average accuracy is 99.09% using the NN(HN=10) model. Enhancing the hidden neuron, the average accuracy achieved is 99.39% using the NN(HN=50) model which is larger than NN(HN=10). It is worth noting that this accuracy aligns with the findings of previous studies conducted on the CHB-MIT dataset shown in Table III, thereby validating the chosen methodology. Moving on to the PHK dataset, similar classification experiments are conducted. The loss curves are shown in Fig. 7 and the accuracy curves are shown in Fig. 8. The results show that the NN models perform well in accurately classifying the data. Specifically, the NN(HNSS=50) model achieves an average accuracy of 91.05%, while the

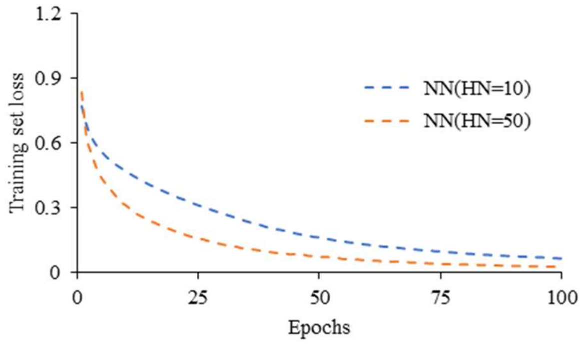

Fig. 7. Training Set Loss Vs. Epoch Curve for a Fold of PHK Dataset.

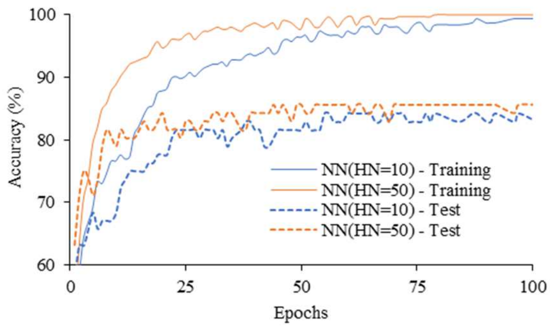

Fig. 8. Accuracy Vs. Epoch Curve for a Fold of PHK Dataset.

TABLE III. ACCURACY COMPARISON FOR PHK DATASET AND CHB-MIT DATASET

Work Ref, Year	Method	Dataset	Accuracy (%)
Pattnaik *et al.* [4], 2022	TQWT + feature extraction + RF	CHB-MIT	93.00
He *et al.* [8], 2022	GAT + BiLSTM	CHB-MIT	98.52
The proposed method	EMD + Feature extraction + NN	CHB-MIT	99.39
		PHK	91.05

NN(HN=10) model achieves an average accuracy of 90.79%. These accuracies indicate that the NN models effectively capture the patterns and features present in the PHK dataset, leading to reliable classification results.

IV. Conclusions

In conclusion, our study presents a method for epileptic seizure detection using EEG analysis. By collecting and processing a new dataset of seizure patients, we successfully apply EMD to extract essential subcomponents from the EEG signals. These subcomponents are then used to derive three specific features, which served as inputs for our NN classifier. The effectiveness of our method is validated through rigorous experiments using both the CHB-MIT dataset and the PHK dataset. On the CHB-MIT dataset, our NN(HN=50) model achieves an outstanding average accuracy of 99.39%, aligning with the findings of previous studies and affirming the reliability of our chosen approach. Furthermore, our NN models continued to demonstrate excellent performance on the PHK dataset. The NN(HN=50) model achieved an average accuracy of 91.05%, while the NN(HN=10) model achieved an average accuracy of 90.79%. Future studies in epileptic seizure detection using EEG analysis can benefit from incorporating CNN and feature selection techniques. Also, classification of epileptic seizure and non-epileptic seizure can benefit the further research.

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